US005400092A

United States Patent [19]

Schepens et al.

[11] Patent Number: 5,400,092

[45] Date of Patent: Mar. 21, 1995

[54] **BINOCULAR OPHTHALMOSCOPE**

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[73] Assignee: **Mira, Inc.**, Waltham, Mass.

[21] Appl. No.: **808,752**

[22] Filed: **Dec. 17, 1991**

[51] **Int. Cl.**[6] **A61B 3/10**

[52] **U.S. Cl.** **351/214**; 351/205; 351/220

[58] **Field of Search** 351/205, 214, 220

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Maxiscope Product Information No date.

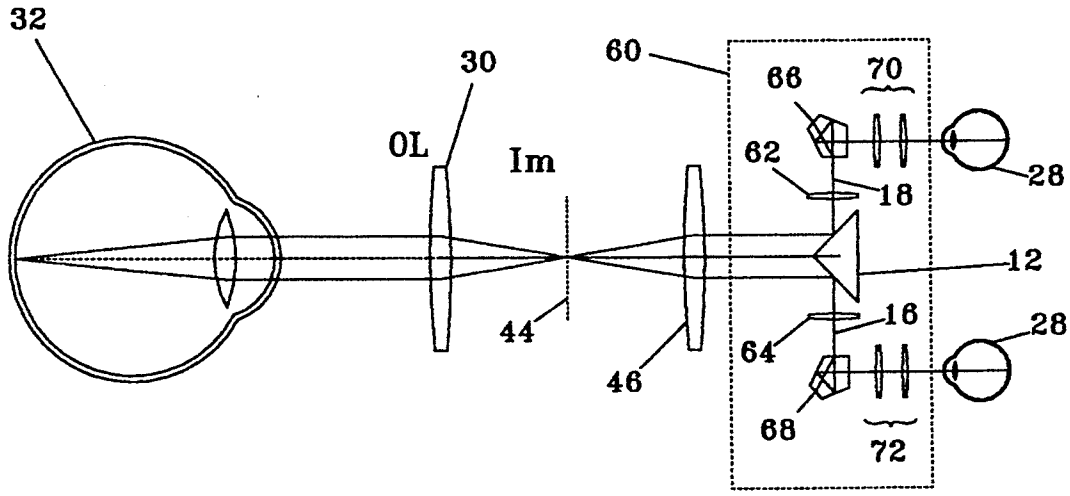
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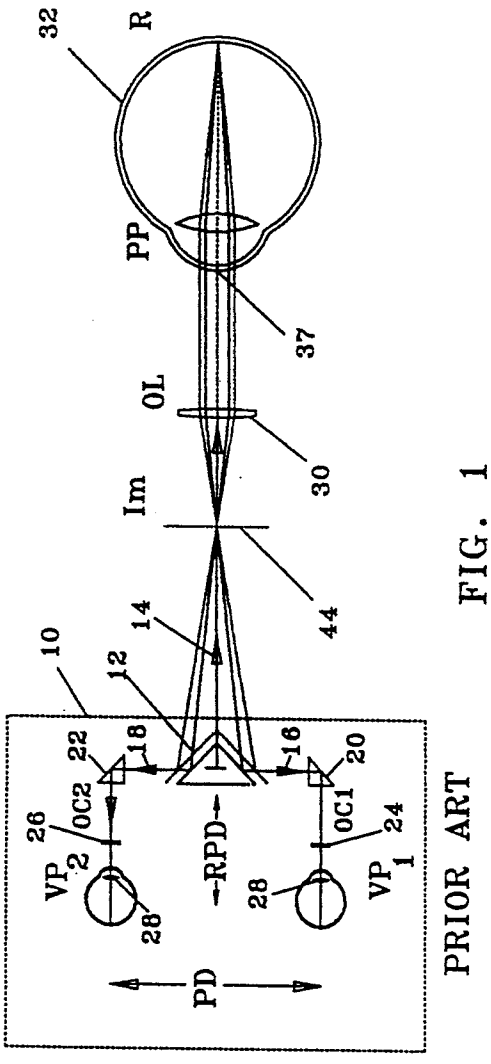
Primary Examiner—William L. Sikes
Assistant Examiner—Hung Xuan Dang
Attorney, Agent, or Firm—Hamilton, Brook, Smith & Reynolds

[57] **ABSTRACT**

A magnifying binocular ophthalmoscope that employs a telecentric ocular lens in front of a viewing mirror assembly in order to provide parallel light rays from the lens to the viewing mirror assembly and from there to two direction changing mirrors that direct the light to the viewer's eyes. Also disclosed is the use of prisms between an objective lens and an ocular lens, the prisms redirecting the optical paths from the viewing mirror assembly and causing the optical path to cross its path within the prism so as to increase the length of the path and invert the image vertically and horizontally. Also disclosed is an assembly for illuminating and directing light that includes first and second stages that are movable along a viewing direction and carry respective first and second mirrors, the second stage being mounted for both movement with the first stage and movement relative to the first stage.

9 Claims, 5 Drawing Sheets





PRIOR ART

FIG. 1

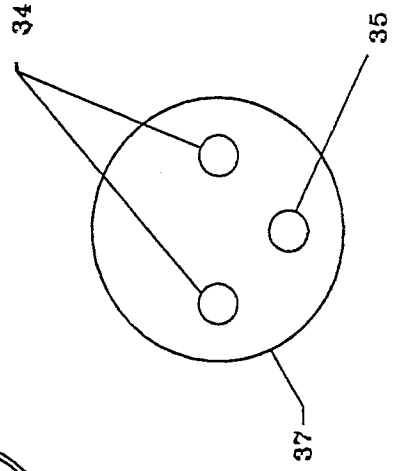
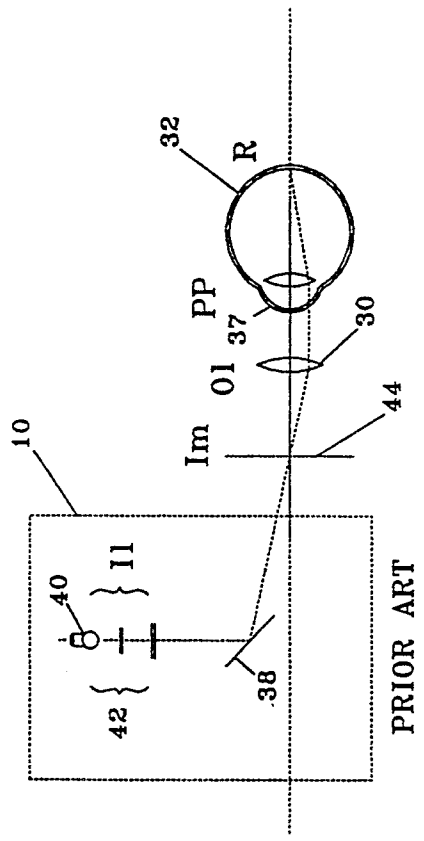


FIG. 2



PRIOR ART

FIG. 3

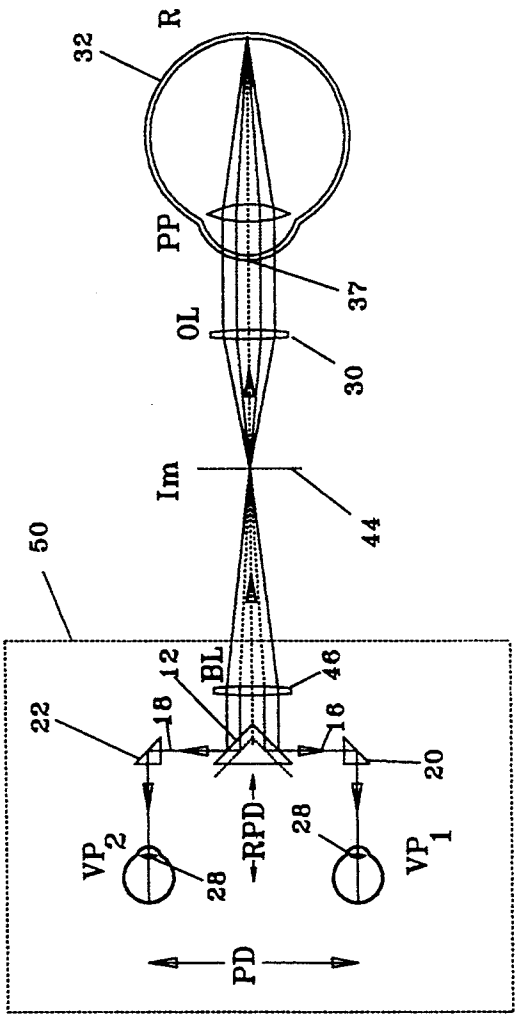

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VP1
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FIG. 4

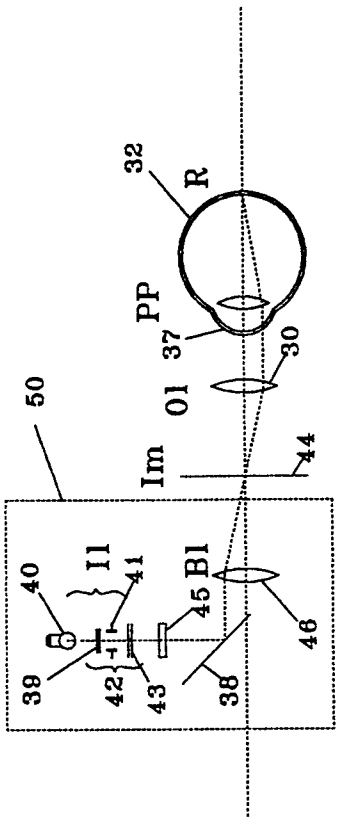

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FIG. 5

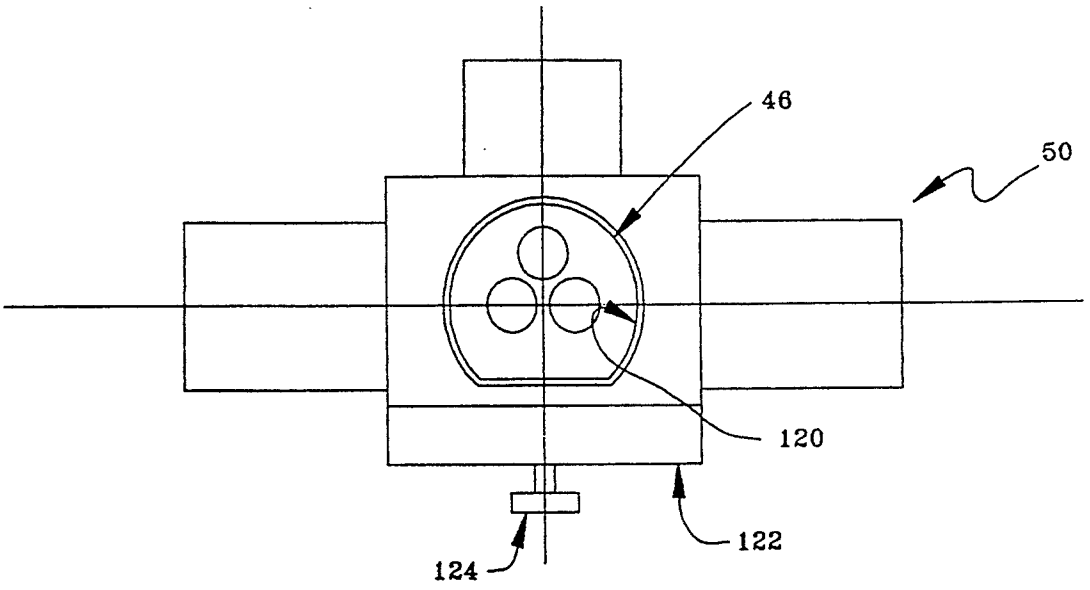

46
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120
122
124

FIG. 6

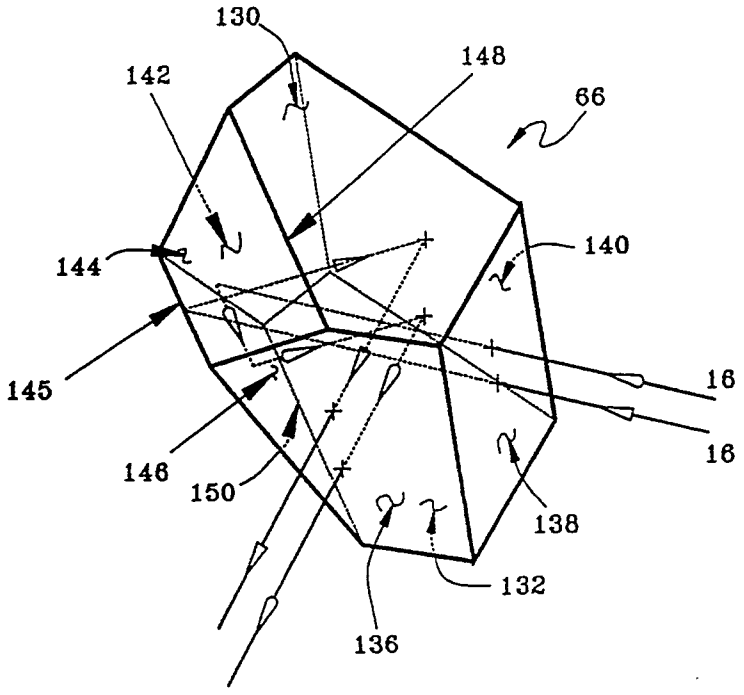

130
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FIG. 8

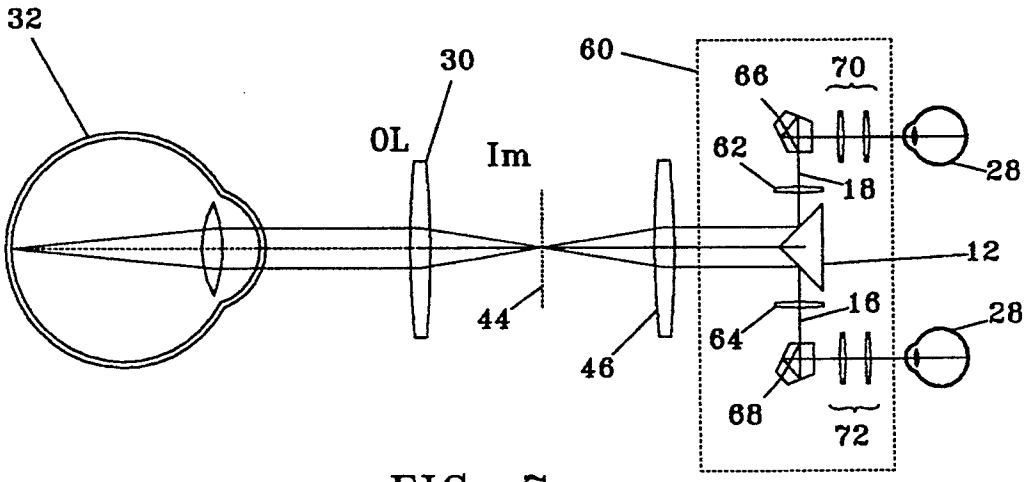

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FIG. 7

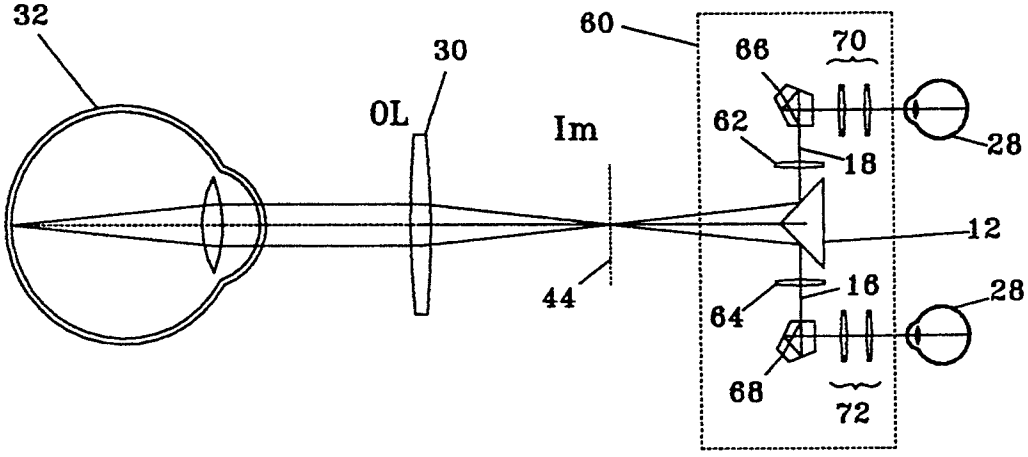

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FIG. 9

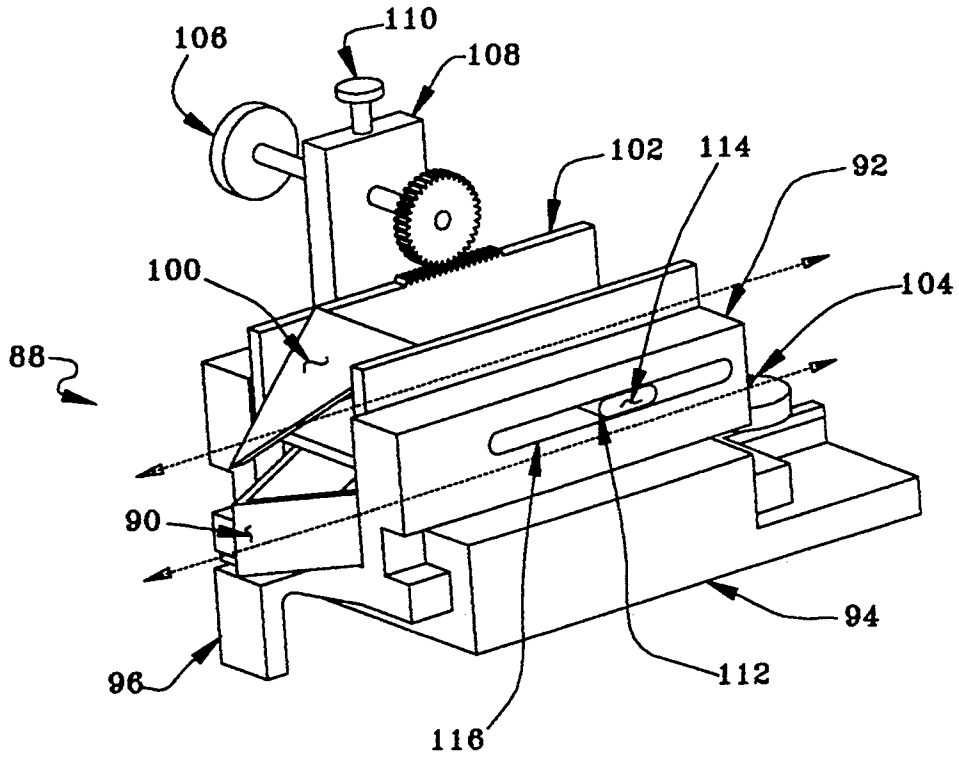


FIGURE 10

BINOCULAR OPHTHALMOSCOPE

The invention relates to magnifying binocular ophthalmoscopes and illuminating systems useful therein.

BACKGROUND OF THE INVENTION

Binocular indirect ophthalmoscopes are used by an ophthalmologist to look into a patient's eye. A unit is strapped onto the ophthalmologist's head, and he holds an ophthalmoscopic lens at arm's length close to the patient's eye, as is discussed in Hovland, K. R. and Schepens, C. L., "Clinical Evaluation of the Small-Pupil Binocular Indirect Ophthalmoscope", *Arch Ophthal,* Vol. 82, October 1969. The unit on the ophthalmologist's head has movable mirrors in order to place images of the observer's two pupils and a light source in the patient's pupil. The head unit accommodates observers having different distances (PD) between their pupils and also has adjustment of the reducing pupil distance (RPD) mechanism in order to adjust the distance between the images of the viewer's two pupils within the patient's pupil. There also typically is movement of a mirror directing light from the illumination source to adjust the distance of the illumination source image from the viewing pupils images within the patient's pupil.

FIGS. 1–3 describe a prior art binocular ophthalmoscope system. Ophthalmologist head unit **10** includes viewing mirror assembly **12** having two reflective surfaces at approximately 45° to the viewing direction along axis **14**. These surfaces provide two optical paths **16, 18** perpendicular to the viewing direction. Mirrors **20, 22** are spaced from each other by the distance PD and redirect the optical paths 90°. The redirected optical paths extend through ocular lenses **24, 26** to the observer's pupils **28**. Ophthalmoscopic lens **30** is held close to the patient's eye **32** and provides an image at image plane **44**. The distance from the viewing pupils **28** to lens **30** is known as the working distance (WD). Viewing mirror assembly **12** is moved forward or backward along the viewing direction of axis **14** (as indicated by the RPD arrows in FIG. **1**) in order to change the distance between the images of the observer's pupils **34**, shown with tho image of tho illumination source **35**, in the patient's pupil **37** in FIG. **2**. When assembly **12** is moved forward, as indicated by dashed lines in FIG. **1**, the images of the viewer's pupils **34** will be further apart in the patient's pupil **37**. Owing to the diverging nature of the light rays from image plane **44**, in some head units **10**, e.g., the S-P Ophthalmoscope sold by MIRA, Inc., Waltham, Mass., the reflective surfaces of mirror assembly **12** are adjusted angularly with forward and backward movements of assembly **12**. In some other head units, e.g., the one shown U.S. Pat. No. 4,684,227, there is no such angular adjustment, with a possible resulting effect on the quality of the stereo image.

Referring to FIG. **3**, showing the illumination system of a prior art binocular ophthalmoscope, mirror **38**, which is separate from mirror assembly **12**, is used to direct light from light source **40** and lens system **42** to the patient's eye **32**. Ophthalmoscopic lens **30** provides an image of light source **40** to the patient's pupil **37**. In some prior art binocular ophthalmoscopes, the illumination mirror is moved independently of the viewing mirror assembly, which typically is located underneath it. In some other prior art binocular ophthalmoscopes,

the illumination mirror is fixedly mounted with respect to the viewing mirror assembly and thus moves with it.

Hubertus U.S. Pat. No. 4,838,678 describes a binocular ophthalmoscope employing plano-convex lens **19** as part of the objective lens system with lenses **28**. The ophthalmoscope also includes eye-piece **34** (including an ocular lens) and Schmidt or Pechan prism **42** to fold the rays of light within it and re-invert the image.

SUMMARY OF THE INVENTION

In one aspect, the invention features, in general, a binocular ophthalmoscope that employs a telecentric ocular lens in front of a viewing mirror assembly in order to provide parallel light rays from the lens to the viewing mirror assembly and from there to two direction changing mirrors that direct the light to the viewer's eyes. By providing parallel rays of light to the viewing mirror assembly, the need for angular adjustment of the mirror assembly with the forward and backward movement is avoided without loss of image quality. This arrangement also provides for advantages in magnification, in that, by having the ocular lens in front, one can decrease the focal length (and increase magnification) without decreasing the working distance. Also, one can get a higher magnification with a wider field by using higher power (short focal length) ophthalmoscopic lenses.

In preferred embodiments, the telecentric ocular lens is removably mounted so as to permit the use of interchangeable lenses having different powers, thus providing different magnification while still maintaining the parallel nature of the light rays. The ophthalmoscope also employs an illumination system that directs light along an optical path through the telecentric ocular lens, providing uniform brightness for different magnifications.

In another aspect, the invention features, in general, a binocular ophthalmoscope employing a pair of objective lenses mounted on opposite sides of the viewing mirror assembly along respective optical paths from the assembly, a pair of prisms mounted outside of the objective lenses, and a pair of ocular lenses between the prisms and the viewer's eyes. The prisms have reflective surfaces that are positioned to redirect the optical paths so as to exit the prisms along the paths that are generally parallel to the viewing direction. The reflective surfaces also cause the optical path to cross its path within the prism so as to increase the length of the path. The reflective surfaces also cause the image to invert vertically and horizontally. This arrangement provides the magnification advantages associated with having an objective lens and an ocular lens and employs the prisms both to change the direction of the optical path at the width of the viewer's eyes and to provide for the necessary inversion of the image created by the use of an additional lens.

In preferred embodiments the prisms are pentaprisms that each have a "roof configuration", i.e., a pair of reflective surfaces at a 90 degree angle with each other in the vertical direction so as to provide a vertical inversion of the image.

In another aspect, the invention features, in general, an assembly for illuminating and directing light from an object. The assembly includes first and second stages that are movable along a viewing direction and carry respective first and second mirrors. (e.g., the first mirror could be in the viewing mirror assembly of a binocular ophthalmoscope and the second mirror could be in the

illumination system.) The second stage is mounted for both movement with the first stage and movement relative to the first stage. This permits the first and second mirrors to desirably be moved as a unit along the direction of sight and also permits for the fine tuning adjustments of the relative positions of the two mirrors by movement of the second mirror relative to the first.

In preferred embodiments there are independent mechanisms to move the two stages, and the second stage is slidably mounted on the first stage such that substantial friction must be overcome to cause relative sliding between the two.

Other advantages and feature is of the invention will be apparent from the following description of the preferred embodiments thereof and from the claims.

DESCRIPTION OF THE PREFERRED EMBODIMENTS

The drawings will be described first.

DRAWINGS

FIG. **1** is a diagram of a prior art binocular ophthalmoscope.

FIG. **2** is diagram showing the positions of images of the observer's pupils and the illumination source in the patient's pupil.

FIG. **3** is a diagram of a prior art illuminating system for a binocular ophthalmoscope.

FIG. **4** is diagram of components of a binocular ophthalmoscope according to the invention.

FIG. **5** is a diagram of the illuminating system of the FIG. **4** ophthalmoscope.

FIG. **6** is an elevation of a unit of the FIG. **4** ophthalmoscope.

FIG. **7** is a diagram of a different embodiment of a binocular ophthalmoscope according to the invention.

FIG. **8** is a perspective view of a pentaprism component of the FIG. **7** ophthalmoscope.

FIG. **9** is a diagram of yet another embodiment of a binocular ophthalmoscope according to the invention.

FIG. **10** is a perspective view of an adjustable mirror assembly which can be used in binocular ophthalmoscopes according to the invention.

STRUCTURE AND OPERATION

Referring to FIG. **4**, a binocular ophthalmoscope according to the invention is shown. The components of the FIG. **4** ophthalmoscope that are in common with those of the prior art ophthalmoscope shown in FIG. **1** have common references numbers. Unit **50**, which is strapped to the ophthalmologist's head, employs viewing mirror assembly **12** and mirrors **20, 22** to direct optical paths to the observer's eyes **28**. Viewing mirror assembly **12** is movable forward and backward to adjust the RPD by a mechanism not shown. Mirrors **20, 22** are likewise movable to adjust the distance PD to accommodate different distances between observer's eyes by a mechanism not shown. These mechanisms are well known in the art. Unit **50** employs a telecentric ocular lens **46**, which provides parallel light rays from it to viewing mirror assembly **12** and along optical paths **16, 18** to viewer's eyes **28**.

Referring to FIG. **5**, light from light source **40** is collimated by condensing lens **39**, passes through diaphragm **41** (located in the focal plane of objective lens **43**), and is focused in optical filter plane **45** (which is ideally located at the same distance from mirror **38** as pupils **28** are from viewing mirror assembly **12**). From plane **45**, the light is reflected by mirror **38** and passes through telecentric ocular lens **46**, forming an image of diaphragm **39** in image plane **44**. After passing through ophthalmoscopic lens **30**, light is focused in the patient pupil plane. By changing ocular lens **46** it is possible to change the illumination diameter in image plane **44** without changing the intensity of light. Such an optical configuration provides uniform image illumination for different magnifications, and reduces the illumination pupil diameter in the patient pupil plane.

Referring to FIG. **6**, lens **46** is removably retained in recess **120** of housing **122** of unit **50** by locking screw **124**, which engages the bottom surface of lens **46**. Three different lenses **46** can be employed. One has power of 6.7 D and a 150 mm focal length. Another has a power of 4 D and a focal length of 250 mm. The third has a power of 10 D and a 100 mm focal length. Also, five different ophthalmoscopic lenses **30** can be used, having powers of 14 D, 20 D, 30 D, 40 D, and 60 D, in the combinations shown in the table below, to provide the indicated magnifications, fields of view in air, patient pupil diameters, and observation distances (OD). The observation distance is the distance between the binocular lens and the aerial image. In the table below ophthalmoscopic lenses **30** are indicated OL.

Optical Parameters Variations Depending on Combinations of Oculars, Back Focal Lengths and Ophthalmoscopic Lenses

Observation Distance	Magnification	Field of View	Pupil Diameter
1. OD = 150 mm			
OL = 14 D (f' = 71 mm)	7*	20°	8 mm
OL = 20 D (f' = 50 mm)	5*	30°	6 mm
OL = 30 D (f' = 33 mm)	3.5*	45°	4 mm
OL = 40 D (f' = 25 mm)	2.5*	60°	3 mm
OL = 60 D (f' = 17 mm)	1.7*	80°	2 mm
2. OD = 250 mm			
OL = 14 D (f' = 71 mm)	5*	25°	6 mm
OL = 20 D (f' = 50 mm)	3*	30°	4 mm
OL = 30 D (f' = 33 mm)	2*	45°	2.5 mm
OL = 40 D (f' = 25 mm)	1.5*	60°	2 mm
OL = 60 D (f' = 17 mm)			
3. OD = 400 mm			
OL = 14 D (f' = 71 mm)	3*	30°	4 mm
OL = 20 D (f' = 50 mm)	2*	40°	2.5 mm
OL = 30 D (f' = 33 mm)	1.5*	45°	2 mm
OL = 40 D (f' = 25 mm)			

In use of the system shown in FIGS. **4–6**, unit **50** is strapped on to the ophthalmologist's head, and lens **30** is held in front of the patient's pupil. Mirrors **20, 22** are moved to accommodate the distance between pupils (PD) for the particular observer, who also adjusts the reducing pupil distance (RPD) mechanism in order to adjust the distance between the images of the viewer's two pupils within the patient's pupil. Mirror **38** (FIG. **5**) is also adjusted forward or backward to adjust the distance of the illumination source image from the 'viewing pupils' images within the patient's pupil. Telecentric ocular lens **46** provides parallel rays of light to the viewing mirror assembly **12** and along paths **16, 18** to the viewing pupils; there is no need for angular adjustment of the mirrors of assembly **12** with the forward and backward movement and no loss of image quality. This arrangement also provides for advantages in magnification, in that by having ocular lens **46** in front, one can decrease the focal length (and increase magnification) without decreasing the working distance. Also, one gets a higher magnification with a wider field by using, for

the same magnification, higher power and wider field ophthalmoscopic lenses. Because the light from the illumination source goes through lens **46**, changing lens **46** to change the magnification changes the illumination diameter **35**, and, as a result, brightens the images uniformly for the range of magnification and field of view.

Referring to FIG. **7**, there is shown a binocular ophthalmoscope system including a unit **60** that is strapped onto the ophthalmologist's head. It includes a binocular lens **46** that is removably mounted, and a movable viewing mirror assembly **12**, to provide two optical paths **16, 18**. Objective lenses **62, 64** are mounted along respective paths **16, 18**, as are pentaprisms **66, 68**. The paths exiting pentaprisms **66, 68** are redirected at 90° and passed through ocular lenses **70, 72** to the viewer's eyes **28**. Each prism **66, 68** has reflective surfaces that are positioned to cause the optical path to cross its path within the prism so as to increase the length of the path, thus accommodating longer focal lengths for the lenses without increasing the size of the head unit. The reflective surfaces also cause the image to invert vertically and horizontally.

Referring to FIG. **8**, prism **66** (prism **68** is identical) has parallel top and bottom surfaces **130, 132**; four vertical surfaces **136, 138, 140, 142**; and two surfaces **144, 146** that are at a 90 degree angle with each other and 45 degree angles with the horizontal. Light enters through surface **138**, is reflected by surfaces **144, 146**, and **140**, and exits through surface **136**. The plane through edge **148** (between surface **130** and **144**) and edge **150** (between surface **146** and surface **132**) makes a 112.5 degree angle with surface **136**. The plane just defined makes a 45 degree angle with surface **140**. Surface **138** makes a 112.5 degree angle with surface **140**. The 112.5 degree and 45 degree angles are common for such pentaprisms, though the surface **144** and **146** at a 90 degree angle with each other and at 45 degree angles with the horizontal are not common. Optical path **16** enters at the center of surface **138** and it is redirected at the junction of surfaces **144, 146** to surface **140** and surface **136**. Optical path **16'** is higher than path **16** at the entrance, but is lower at the exit, as it is redirected from surface **144** vertically downward to surface **146** before being redirected to surface **140**. Similarly, a path under center path **16** at surface **138** exits above it at surface **136**. A path to the right of path **16** at surface **138** is on the left side coming out at surface **136**. The image coming out is thus inverted both vertically and horizontally from that going in, unlike a typical pentaprism (which has a planar reflective surface at the plane through edges **148, 150** instead of surfaces **144, 146**), which inverts the image horizontally but not vertically.

The arrangement of FIG. **7** provides the higher magnification advantages associated with having an objective lens and an ocular lens and employs prisms **66, 68** to perform the dual functions of changing the direction of the optical path at the width of the viewer's eyes and providing for the necessary inversion of the image created by the use of an additional lens. In addition, prisms **66, 68** achieve these goals with only three reflections, thus limiting the losses associated with reflections. A further advantage of prisms **66, 68** is that they are insensitive to slight angular variation with respect to the optical paths **16, 18**, so that some variation of or lack of precision in orientation of prisms **66, 68** during mounting will not detrimentally affect operation.

As with the FIGS. **4** and **5** embodiment, the FIGS. **7** and **8** embodiment can be used with the same three different lens **46** and with the same five different ophthalmoscopic lenses **30** in the combinations shown in the table below, to provide the indicated magnifications, fields of view in air, patient pupil diameters, and observation distances (OD).

Optical Parameters Variations Depending on Combinations of Oculars, Back Focal Lengths and Opthalmoscopic Lenses

Observation Distance	Magnification	Field of View	Pupil Diameter
1. OD = 150 mm			
OL = 14 D (f' = 71 mm)	20*	20°	10 mm
OL = 20 D (f' = 50 mm)	15*	30°	7 mm
OL = 30 D (f' = 33 mm)	10*	45°	4 mm
OL = 40 D (f' = 25 mm)	7.5*	60°	3.5 mm
OL = 60 D (f' = 17 mm)	5*	80°	2.3 mm
2. OD = 250 mm			
OL = 14 D (f' = 71 mm)	12*	25°	6 mm
OL = 20 D (f' = 50 mm)	9*	30°	4 mm
OL = 30 D (f' = 33 mm)	5*	45°	2.5 mm
OL = 40 D (f' = 25 mm)	4.5*	60°	2 mm
OL = 60 D (f' = 17 mm)	3*	80°	1.3 mm
3. OD = 400 mm			
OL = 14 D (f' = 71 mm)	7.5*	25°	4 mm
OL = 20 D (f' = 50 mm)	5.6*	35°	2.5 mm
OL = 30 D (f' = 33 mm)	3.7*	45°	1.6 mm
OL = 40 D (f' = 25 mm)	2.8*	60°	1.25 mm

FIG. **9** describes a system that is similar to FIG. **7** except that it does not employ binocular lens **46**. In this embodiment, changes in RPD would require changes in the working distance.

Referring to FIG. **10**, there is shown optical assembly **88** that can be used in the binocular ophthalmoscopes described herein or in other applications. Viewing mirror assembly **90** is mounted on first stage **92** for sliding movement along the viewing direction axis with respect to base **94** by manually pulling slide adjustment tab **96**. Illumination mirror **100** is mounted on second stage **102** for sliding along a parallel axis, either with first stage **92** or relative to first stage **92**. First stage **92** can be locked with respect to base **94** by tightening locking screw **104**. Second stage **102** can be moved relative to base **94** by rotation of knob **106**, mounted on support **108**, attached to base **94**. Locking screw **110** can be tightened to lock knob **106**. Compression slide **112** is mounted on second stage **102** and has ends **114** in slots **116** of first stage **92**. (Only one end **114** and one slot **116** are shown in FIG. **10**.) Ends **114** have a thickness slightly greater than the height of slot **116**, and provide a sufficiently high coefficient of friction so as to cause the two stages to move with each other unless one of the locking screws are tightened; in this case, slide **112** slides within slot **116**. If locking screw **104** is tightened, and locking screw **110** is not, rotation of knob **106** causes only second stage **102** to move. If locking screw **110** is tightened, and locking screw **104** is not, pushing or pulling of tab **96** causes only first stage **92** to move. If both locking screws are not tightened, the two stages will move together when either tab **96** or knob **106** are activated. This permits viewing mirror assembly **90** and illumination mirror **100** to desirably be moved as a unit along the direction of sight and also permits for the fine tuning adjustments of the relative positions of the two sets of mirrors **90, 100** and the independent movement of the two sets of mirrors **90, 100**.

Other embodiments of the invention are within the scope of the claims.

What is claimed is:

1. A head mounted binocular ophthalmoscope comprising:
a support structure having a strap to mount the support structure on a viewer's head,
a viewing mirror assembly centrally mounted for movement along a viewing direction on said support structure, said viewing mirror assembly having mirrors mounted to provide two optical paths generally perpendicular to said viewing direction and extending in opposite directions from said viewing mirror assembly,
objective lenses mounted on said support structure on opposite sides of said viewing mirror assembly along the two respective optical paths,
spaced prisms mounted on said support structure on opposite sides of said viewing mirror assembly along respective said optical paths outside of said objective lenses and spaced from each other by the distance between the viewer's eyes, said prisms having surfaces providing reflections to redirect said optical paths entering said prism so as to exit said prisms along paths that are generally parallel to said viewing direction and each other and spaced from each other by said distance equal to the spacing between the viewer's eyes, said reflective surfaces causing said optical path to cross its path within said prism so as to increase the length of the path, said reflective surfaces also causing the image to be inverted vertically and horizontally, and
ocular lenses in respective said optical paths exiting said prisms.

2. The ophthalmoscope of claim 1 wherein said prism is a pentaprism with a pair of reflective surfaces at 90 degree angle with each other in a vertical cut so as to provide a vertical inversion of the image.

3. The ophthalmoscope of claim 2 further comprising a telecentric ocular lens in front of said mirror assembly providing parallel light rays from it to said viewing mirror assembly.

4. The ophthalmoscope of claim 3 wherein said telecentric ocular lens is removably mounted so as to permit use of interchangeable lenses having different power.

5. The ophthalmoscope of claim 4 further comprising an illumination system that directs light along an optical path through said telecentric ocular lens.

6. A binocular ophthalmoscope for directing illuminating light to an object being viewed and directing light from the illuminated object comprising;
a first stage that is movable along a viewing direction,
a first mirror assembly that is carried on said first stage and oriented to direct light from said illuminated object along a first path and a second path which are transverse to said viewing direction,
a pair of mirrors or prisms mounted to reflect light along the first path and second path,
a second stage that is movable along said viewing direction and is adjustably mounted relative to said first stage to selectively have movement with said first stage or movement relative to said first stage and the first mirror assembly, and
a second mirror that is carried on said second stage and oriented to direct light from a source to said object, said light traveling from said source along a second path that is transverse to the viewing direction, the second mirror being movable with the first stage along the viewing direction, or alternatively, being movable relative to the first stage.

7. The assembly of claim 6 wherein said first mirror assembly receives light from the object through an ocular lens mounted on a support structure relative to the first mirror assembly.

8. The optical assembly of claim 6, further comprising spaced prisms for directing light from said viewing direction to an optical path 90° to said viewing direction, said spaced prisms being spaced from each other by the distance between the viewer's eyes, said spaced prisms having surfaces providing reflections to redirect said optical paths entering said prism so as to exit said spaced prisms along paths that are generally parallel to said viewing direction and each other and spaced from each other by said distance equal to the spacing between the viewer's eyes, said reflective surfaces causing said optical path to cross its path within said prism so as to increase the length of the path, said reflective surfaces also causing the image to be inverted vertically and horizontally.

9. A binocular ophthalmoscope comprising:
a support structure;
a viewing mirror assembly centrally mounted for movement along a viewing direction on said support structure, said viewing mirror assembly having mirror mounted to provide two optical paths generally perpendicular to said viewing direction and extending in opposite directions from said viewing mirror assembly;
a pair of spaced prisms mounted on said support structure on opposite sides of said viewing mirror assembly along respective said optical paths and spaced from each other by the distance between the viewer's eyes, said spaced prisms being oriented to redirect said optical paths from said assembly so as to be generally parallel to said viewing direction and each other and spaced from each other by said distance equal to the spacing between the viewer's eyes;
an ocular lens in front of said viewing mirror assembly providing parallel light rays from the ocular lens to said viewing mirror assembly and along said paths to the viewer's eyes whereby a telecentric lens is formed, the spaced prisms being mounted on said support structure on opposite sides of said assembly along respective said optical paths and spaced from each other by the distance between the viewer's eyes, said prisms each having surfaces providing a plurality of reflections to redirect said optical paths entering said prism so a to exit said prism along paths that are generally parallel to said viewing direction and each other and spaced from each other by said distance equal to the spacing between the viewer's eyes, said reflective surfaces causing said optical path to cross its paths within said prism so as to increase the length of the path, said reflective surfaces also causing the image to be inverted vertically and horizontally.

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